



Original Contribution

Ultrasound-guided paravertebral block with liposomal bupivacaine vs. non-liposomal bupivacaine for postoperative pain control after the Nuss procedure in children: A randomized clinical trial

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HIGHLIGHTS

- Paravertebral block with LB reduced opioid use in children after Nuss procedure compared to nLB.
- Paravertebral block with LB reduced incidence of rebound pain in children after the Nuss procedure compared to nLB.
- Paravertebral block with LB is recommended as part of multimodal analgesia during Nuss procedure in children.
- Despite multimodal analgesia, some children post-Nuss procedure still experience moderate-severe pain on movement.

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ABSTRACT

Study objective: Minimally invasive repair of pectus excavatum (Nuss procedure) is associated with significant postoperative pain and high-dose opioid consumption. Liposomal bupivacaine (LB), as an ultra-long-acting local anesthetic, has been extensively studied for perioperative analgesia. However, data regarding the effects of LB on postoperative pain in children undergoing the Nuss procedure remain limited. The aim of this study was to evaluate the ability of LB to reduce postoperative opioid use among children undergoing the Nuss procedure.

Design: Randomized controlled trial.

Setting: Operating room.

Patients: Children aged 6–18 years who were scheduled for elective Nuss procedures.

Interventions: Patients were randomly assigned to receive ultrasound-guided paravertebral block with either LB or non-liposomal bupivacaine (nLB) after anesthesia induction.

Measurements: The primary outcome was the consumption of morphine equivalent during the first 72 h after surgery. The secondary endpoints included the maximum Numerical Rating Scale (NRS) pain score within 72 h, incidence of moderate-to-severe pain within 72 h, incidence of rebound pain within 72 h, NRS sleep score at 3 days, incidence of postoperative nausea and vomiting, length of hospital stay after surgery, and time to first ambulation after the operation.

Main results: A total of 109 subjects were included in the intention-to-treat analysis. Within 72 h after surgery, the morphine equivalent consumption in the LB group was reduced by about 23 % compared with the nLB group (median difference: 23.1 mg; $P = 0.023$). The incidence of rebound pain was significantly lower in the LB group than in the nLB group (13.0 % [7/54] vs. 29.1 % [16/55], relative risk 0.46; 95 % CI: 0.20–0.96; $P = 0.039$). Subjective sleep quality was better in the LB group than in the nLB group on the second and third nights after surgery (median difference: –1 point; $P = 0.005$ and $P = 0.016$). The incidence of adverse events was similar in both groups, and no patients experienced severe adverse events during the study period.

Conclusions: Paravertebral block with LB reduces opioid use (a ~ 23 % reduction) and the incidence of rebound pain in children after the Nuss procedure compared to nLB group.

Trial registration: www.chictr.org.cn (Registration number ChiCTR2400085350, Registration date June 5, 2024).

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1. Introduction

Pectus excavatum (PE) is a common anterior chest wall deformity with an incidence of approximately 1 in 1000 [1]. Moderate to severe PE is treated mainly by surgery, with the Nuss procedure, a minimally invasive surgery for PE repair, as the first choice [2,3]. Although the Nuss procedure is minimally invasive, postoperative pain management is a significant challenge [4]. In addition to incision pain, patients also experience muscular and neuropathic pain associated with stretching of the intercostal muscle tissue, which persists for an extended period after the procedure is complete [5]. Therefore, the Nuss procedure is associated with significant postoperative pain and high-dose opioid consumption [1,4,6]. For this type of surgery, the predominant postoperative pain management strategy is multimodal analgesia, such as non-opioid medications and regional/neuraxial techniques [6,7]. Unfortunately, despite the use of multimodal analgesia, patients who undergo the Nuss procedure remain dissatisfied with pain management, and opioid consumption remains high [6,7]. Moreover, owing to the limited analgesic duration of hydrochloric local anesthetics, the incidence of postoperative rebound pain is greater in patients undergoing this type of surgery [8,9].

Liposomal bupivacaine (LB) is an ultra-long-acting local anesthetic that has been used for perioperative analgesia in clinical practice for several years [10]. Previous studies have reported potential advantages of perineural LB over conventional non-liposomal bupivacaine (nLB) in postoperative pain management and opioid reduction [11–14]. However, this finding remains controversial as evidenced by a systematic review by Ilfeld BM et al., which concluded that LB demonstrates limited clinical benefits in postoperative analgesia [15]. Moreover, studies have shown that LB for intercostal nerve block also shows some advantages over nLB in pain management after Nuss procedure [16,17]. However, none of these studies were designed as prospective randomized controlled clinical trials, and their outcomes carry a substantial risk of bias, necessitating higher-level evidence to substantiate these preliminary findings. Notably, paravertebral nerve block is a safe and effective measure for postoperative analgesia in Nuss procedure [18,19]. Therefore, the current study aimed to evaluate the effect of paravertebral block combined with LB or nLB on pain control after Nuss procedure in a prospective controlled clinical trial.

In this study, the primary hypothesis was that paravertebral nerve block with LB may reduce morphine equivalent consumption within 72 h after the Nuss procedure in children. Secondary endpoints included the maximum Numerical Rating Scale (NRS) pain score within 72 h, incidence of moderate-to-severe pain within 72 h, incidence of rebound pain within 72 h, NRS sleep score within 3 days, incidence of postoperative nausea and vomiting, length of hospital stay after surgery, and time of first ambulation after the operation.

2. Methods

2.1. Study design and population

This prospective, single-center RCT was conducted at the Departments of Anesthesiology and Cardiothoracic Surgery at the Affiliated Children's Hospital of Xiangya School of Medicine from June to November 2024. Ethical approval for this study was provided by the Ethics Committee of Hunan Children's Hospital, Changsha, China (Chairperson Prof Zhenghui Xiao; Ethics approval number: HCHLL-2024-17), in January 2024. Before recruitment, the trial was registered at the Chinese Clinical Trial Registry website (ChiCTR2400085350). All parents/guardians provided written informed consent for the participation of their children in this study, and assent from the patient was sought when applicable.

2.2. Participants

Children aged 6–18 years who were scheduled for elective Nuss procedures at the Affiliated Children's Hospital of Xiangya School of Medicine were recruited. Patients who met any of the following criteria were excluded: congenital heart disease, arrhythmia, or other serious cardiovascular diseases; scoliosis deformities; allergies to local anesthetics or contraindications for nerve block; a history of immunodeficiency diseases; the use of immunosuppressants within 30 days before surgery; a history of mental illness (such as schizophrenia, depression, etc.) a communication barrier; a history of chronic or visceral pain; abnormal liver or kidney function; or other conditions that were considered unsuitable for study participation.

2.3. Randomization and blinding

The participants were randomly assigned to either the LB group or the nLB group at a 1:1 ratio according to a computer-generated random number table. The allocation information was kept in opaque envelopes and was not revealed until the investigators administered the intervention after anesthesia induction. In this study, the anesthesiologist team, the surgeon team, and the nursing team responsible for perioperative management of the subjects were unaware of the grouping. Moreover, the outcome assessors, the researchers responsible for data processing and statistics, other health care personnel, and patients were also blinded to the treatment allocation.

2.4. Anesthesia and intervention

After entering the operating room, the subject underwent routine monitoring, including electrocardiography, pulse oximetry, and noninvasive blood pressure. Venous access was established under local anesthesia for all the subjects. Dexamethasone (5–10 mg) was administered before anesthesia induction to prevent postoperative nausea and vomiting. Anesthesia induction was performed with midazolam, sufentanil, propofol, cisatracurium, or rocuronium.

After anesthesia induction, the subjects underwent ultrasound-guided paravertebral nerve block in the left decubitus position. Paravertebral blocks were performed on both sides of the 4th to 8th thoracic vertebrae, with two puncture block points on each side (depending on the location and extent of the sternal indentation). For subjects in the LB group, 4 to 8 ml of 0.665 % LB (266 mg/20 ml of LB diluted with normal saline to 40 ml) was administered at each point of the paravertebral block. For subjects in the nLB group, 4 to 8 ml of 0.25 % bupivacaine was administered at each point of the paravertebral block. The total volume of the local anesthetics in all the subjects did not exceed 40 ml, and the LB dose did not exceed 4 mg kg^{-1} . Notably, all paravertebral nerve blocks were performed by the same anesthesiologist, who was not involved in anesthesia management, outcome assessment, or data processing. The anesthesiologists responsible for intraoperative management were blinded to the group assignments because they were not involved in the paravertebral block process. Anesthesia was maintained with propofol infusion and sevoflurane inhalation. Remifentanyl infusion OR single-dose sufentanil/oxycodone was used for intraoperative analgesia. Cisatracurium or rocuronium was used for neuromuscular block. The bispectral index (BIS) values were maintained at 40–60 by adjusting the sevoflurane inhalation concentration or the propofol infusion rate. Before the surgical incision was made, radial catheterization and invasive blood pressure monitoring were performed. Moreover, all the procedures were performed by the same surgical team, which was blinded to the group assignments.

After completion of the procedure, the subjects were transferred to the post-anesthesia care unit (PACU). Before tracheal extubation, the subjects were routinely given neostigmine or sugammadex for antagonism of the neuromuscular blockade and ondansetron for prevention of postoperative nausea and vomiting. All the subjects began receiving

patient-controlled intravenous analgesia (PCIA) after extubation. PCIA was established with $3 \mu\text{g kg}^{-1}$ sufentanil (up to $200 \mu\text{g}$), which was diluted with normal saline to 150 ml. PCIA was programmed to deliver 3 ml boluses with a lockout interval of 10 min and a background infusion rate of 0.5 ml h^{-1} . The duration of PCIA was 72 h after surgery. Subjects who exhausted PCIA analgesics within 72 h or still required PCIA after 72 h were given $3 \mu\text{g kg}^{-1}$ sufentanil (up to $200 \mu\text{g}$) in 150 ml of normal saline. When the modified Aldrete score was ≥ 9 , the subject was transferred to the ward. Upon returning to the ward, the patient routinely received intravenous ibuprofen 10 mg kg^{-1} three times daily until postoperative day 3 for adjunctive analgesia. Rescue analgesia, including intravenous or oral opioids (such as hydromorphone, oxycodone, or morphine), was administered if needed; the goal was to maintain an NRS score ≤ 3 on an 11-point scale where 0 = no pain and 10 = worst pain [20]. Notably, the NRS assessment of the patients and rescue analgesia if necessary were performed by the same anesthesiologist who was unaware of the group assignments.

2.5. Clinical outcomes and assessments

The primary outcome was the consumption of morphine equivalents during the first 72 h after surgery. All opioids, including sufentanil in PCIA and intravenous or oral opioids used for rescue analgesia, were converted to intravenous morphine equivalents. The secondary outcomes included maximum and minimum pain intensity within 3 days after surgery, incidence of moderate-to-severe pain within 3 days, incidence of rebound pain, sleep quality within 3 days, incidence of postoperative nausea and vomiting, and time of first ambulation after the operation. Pain intensity was assessed via the NRS, with the maximum pain intensity score defined as pain with activity or the highest pain score experienced by the patient if they were unable to determine their pain level with activity; the minimum pain intensity score was defined as pain at rest, either lying down or sitting [12]. Subjective sleep quality was also evaluated with the NRS (an 11-point scale where 0 = best possible sleep and 10 = worst possible sleep) [21]. Moderate to severe pain was defined as NRS pain score ≥ 4 at any point in time. Rebound pain was defined as the transition from well-controlled pain (NRS ≤ 3) while the block is working to severe pain (NRS ≥ 7) within 72 h of block performance [22].

Safety data were monitored from the beginning of the paravertebral nerve block until discharge. We monitored the occurrence of hypotension (more than a 30 % decrease in mean arterial pressure from baseline for >5 min), hypertension (more than a 30 % increase in mean arterial pressure from baseline for >5 min), bradycardia (more than a 20 % decrease in the baseline heart rate for >30 s), and tachycardia (more than a 20 % increase in the baseline heart rate for >30 s) [23]. Moreover, we monitored the occurrence of allergic reactions, numbness in the limb or skin, dizziness, and headache.

2.6. Statistical analysis

An analysis of previous data revealed that the mean morphine equivalent consumption after the Nuss procedure was 101 mg, with a standard deviation of 30. In our preliminary study, the mean morphine equivalent consumption after Nuss surgery decreased by 17 mg with the use of LB for paravertebral block. Forty-nine subjects were required for each group to provide 80 % power at an alpha value of 0.05. To allow for withdrawal or loss of patients for other reasons, 110 subjects were planned to be enrolled in this study.

For the primary outcome, we performed not only intention-to-treat (ITT) analysis but also per-protocol (PP) analysis. ITT analyses were performed for all the other outcomes.

Data distribution was evaluated via the Shapiro–Wilk and Anderson–Darling tests. For the perioperative data, continuous variables with a normal distribution are expressed as the mean (standard deviation, SD) and were analyzed via independent-sample *t*-tests; those with a

nonnormal distribution are expressed as the median (interquartile range, IQR) and were analyzed via Mann–Whitney *U* tests. The categorical variables were analyzed with χ^2 tests or Fisher's exact tests. For primary and secondary outcome data, continuous variables with non-normal distributions and ordinal variables were analyzed via the Mann–Whitney *U* test; categorical variables were analyzed via the χ^2 test or Fisher's exact test; and between-group differences are expressed as relative risks or median difference and 95 % CIs. A *P* value <0.05 was considered to indicate statistical significance. All the data were analyzed via SPSS, version 25.0 (IBM Corp).

3. Results

3.1. Study population

A total of 125 patients scheduled to undergo the elective Nuss procedure were screened for eligibility from June to November 2024. Of these, 10 patients did not meet the inclusion criteria, and 5 patients declined to participate in the trial. Ultimately, 110 patients were enrolled and randomly assigned to either the LB ($n = 55$) or nLB ($n = 55$) group. During the study period, one patient withdrew consent after surgery, and 10 patients used PCIA for less than 72 h. Finally, 109 patients were included in the ITT analysis, and 99 patients were included in the PP analysis. The flow diagram of the participant inclusion process is shown in the Fig. 1, and the demographic characteristics are detailed in Table 1. Intraoperative data and postoperative data, including durations of anesthesia and surgery, number of steel plates, medications used during anesthesia, fluid intake and output, hemodynamic data, PACU discharge time and length of hospital stay after surgery, did not differ between the two groups (Table 2).

3.2. Outcomes

The postoperative morphine equivalent consumption was significantly lower in the LB group than in the nLB group within 48 h (median difference: 13.6 mg; 95 % CI: 0.04–27.9; $P = 0.048$) and within 72 h (median difference: 23.1 mg; 95 % CI: 2.4–37.1; $P = 0.023$) (Table 3). The PP analysis revealed similar results: within 48 h, the median difference between the two groups was 12.2 mg ($P = 0.043$); within 72 h, the median difference between the two groups was 23.4 mg ($P = 0.018$) (Table 3). Overall, the postoperative morphine equivalent consumption was reduced by approximately 23 % in the LB group relative to the nLB group within 72 h after surgery.

There was no difference in the median maximum NRS pain score at 1 or 2 days after surgery between the two groups. Although there was a statistical difference in the maximum NRS pain score between the two groups on the third day after surgery, the difference was not clinically significant (median difference: 0 points; 95 % CI: 0–1; $P = 0.036$). The incidence of rebound pain was significantly lower in the LB group than in the nLB group (13.0 % [7/54] vs. 29.1 % [16/55], relative risk 0.45; 95 % CI: 0.20–0.96; $P = 0.039$; Table 3). Subjective sleep quality was better in the LB group than in the nLB group on the second and third nights after surgery (median difference: –1 point; 95 % CI: 0–1; $P = 0.005$ and $P = 0.016$; Table 3). Other secondary outcomes, including moderate-to-severe pain, postoperative nausea and vomiting, and the time of first ambulation after the operation, did not differ between the two groups.

In terms of safety data, the incidence of adverse events was similar in both groups, and no patients experienced severe adverse events during the study period (Table 4).

4. Discussion

In children who underwent the Nuss procedure, compared with paravertebral nerve block with nLB, paravertebral nerve block with LB reduced morphine equivalent consumption within 72 h by

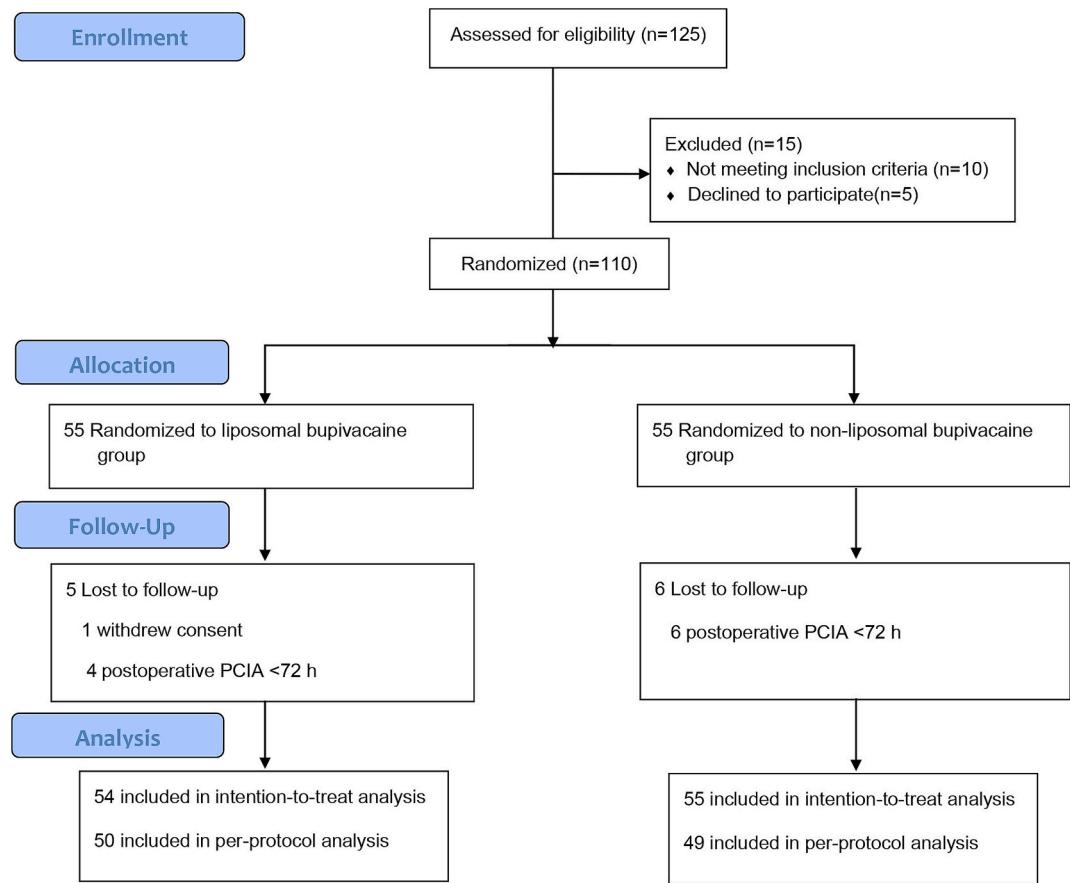


Fig. 1. Study flow chart. PCIA, patient-controlled i.v. analgesia.

Table 1
Baseline characteristics of study participants in the liposomal bupivacaine (LB) and non-liposomal bupivacaine (nLB) groups.

Characteristic	LB(n = 54)	nLB(n = 55)	P-value
Age, mean(SD), y	13.0(2.3)	13.4(2.3)	0.409
Sex			
Male	44(81.5)	43(78.2)	0.668
Female	10(18.5)	12(21.8)	0.668
Weight, mean(SD), kg	41.9(9.8)	43.8(11.1)	0.375
BMI, mean(SD)	16.5(2.0)	16.7(2.3)	0.685
ASA classification			
I	41(75.9)	42(76.4)	0.957
II	13(24.1)	13(23.6)	0.957
History of surgical procedures	13(24.1)	12(21.8)	0.779
NRS of pain, at rest, median(IQR)	0(0–0)	0(0–0)	0.889
NRS of pain, with movement, median(IQR)	1(0–1)	1(0–1)	0.236
NRS of sleep, median(IQR)	1(0–2)	1(0–1.5)	0.203
Basal mean arterial pressure, median(IQR),mmHg	77.6 (69.5–82.3)	75.6 (69.3–82.2)	0.480
Basal heart rate, mean(SD), min ⁻¹	83.6(6.3)	82.7(7.3)	0.523
Haller index, mean(SD)	4.1(0.42)	4.1(0.41)	0.704
Exertion dyspnoea	5(9.3)	7(12.7)	0.836

Data expressed as n(%) unless otherwise stated. BMI, body mass index (calculated as weight in kilograms divided by height in meters squared); ASA, american society of anesthesiologists physical status classification; NRS, numerical rating scale; SD, standard deviation; IQR, inter-quartile range.

approximately 23 mg (a ~ 23 % reduction). Moreover, paravertebral nerve block with LB reduced the incidence of rebound pain within 72 h and improved subjective sleep quality on the second and third days after surgery without increasing associated adverse events. Hence, LB combined with nerve block can be recommended as a component of multimodal analgesia for Nuss surgery in children.

The Nuss procedure is widely used in pediatric PE repair [1,2,24]. However, postoperative pain management is a significant challenge because of the high intensity and long duration of pain [4,5,17]. Poor postoperative analgesia not only increases the anxiety of patients and their families but also extends the postoperative recovery time of patients [25]. Thoracic epidural analgesia and intercostal cryoablation techniques are considered effective measures for treating pain after the Nuss procedure [26,27]. These postoperative analgesia techniques demonstrate distinct advantages while presenting certain limitations. Thoracic epidural analgesia can provide excellent analgesia following Nuss procedure [25,27]. However, this method presents several challenges: (1) technical difficulties in catheter management, particularly in pediatric patients; (2) potential neurological complications; and (3) possible hemodynamic instability secondary to sympathetic blockade [25,28,29]. Intercostal cryoablation for post-Nuss procedure analgesia has been shown to substantially decrease both postoperative opioid consumption and hospital length of stay [25,26]. Notably, intercostal cryoablation can provide exceptionally prolonged analgesic effects. Nevertheless, this technique entails several limitations, including increased procedural costs, prolonged operating room time, and the requirement for surgeons to master a relatively novel technique. Additionally, while long-term outcome data remain limited, cryoablation may carry a risk of persistent neuropathic pain [25]. An analysis of patient satisfaction reported that the use of patient-controlled analgesia (PCA) with opioid medications could achieve satisfactory analgesic

Table 2

Intraoperative and postoperative data between the liposomal bupivacaine (LB) and non-liposomal bupivacaine (nLB) groups.

Characteristic	LB(n = 54)	nLB(n = 55)	P value
Duration of operation, mean (SD), min	62.0(7.1)	60.9(6.5)	0.391
Duration of anesthesia, mean (SD), min	139.0(15.2)	137.3(16.8)	0.587
Number of steel plates			
1	42(77.8)	44(80.0)	0.776
2	12(22.2)	11(20.0)	
Medications during anesthesia			
Use of midazolam	7(13.0)	8(14.5)	0.810
Midazolam, median(IQR), mg	2(1–2)	2(1–2)	>0.999
Propofol, median(IQR), mg	233.0	222.0	0.457
	(201.3–267.0)	(198.0–257.0)	
Use of Dexmedetomidine	22(40.7)	21(38.2)	0.785
Dexmedetomidine, median (IQR), ug	50.0	43.0	0.488
	(38.8–56.5)	(38.5–56.0)	
Sufentanil, median(IQR), ug	30.0	25.0	0.134
	(25.0–35.0)	(25.0–35.0)	
Use of remifentanyl	13(24.1)	11(20.0)	0.608
Remifentanyl, mean(SD), mg	0.44(0.15)	0.48(0.19)	0.531
Oxycodone, median(IQR), mg	5.0(4.0–5.0)	5.0(4.0–6.0)	0.459
Dexamethasone, median(IQR), mg	5.0(5.0–5.0)	5.0(5.0–5.0)	>0.999
Intraoperative morphine equivalent, median(IQR), mg	41.4	39.0	0.594
	(32.5–62.0)	(32.5–47.5)	
Infusion quantity, mean (SD), mL	634.8(137.8)	618.4(144.6)	0.545
Crystalloid median(IQR), mL	600.0	600.0	0.466
	(500.0–700.0)	(500.0–700.0)	
Use of artificial colloid	8(14.8)	10(18.2)	0.636
Artificial colloid, median (IQR), mL	175.0	150.0	0.376
	(150.0–275.0)	(150.0–200.0)	
Estimated blood loss, median (IQR), mL	30.0	25.0	0.338
	(25.0–35.0)	(25.0–35.0)	
Urine output, mean(SD), mL	134.4(42.4)	137.5(39.9)	0.686
Mean arterial pressure, median (IQR), mmHg	70.5	71.0	0.835
	(65.8–79.3)	(68.0–80.0)	
Heart rate, median(IQR), min ⁻¹	80.0	79.0	0.550
	(74.8–83.0)	(70.0–85.0)	
PACU discharge time, median (IQR), min	50.0	55.0	0.072
	(45.0–60.0)	(50.0–65.0)	
Ibuprofen, mean(SD), g	3.8(0.89)	3.9(0.95)	0.657
Length of hospital stay after surgery, median(IQR), day	4.0(3.0–5.0)	4.0(3.0–4.0)	0.142

Data expressed as n(%) unless otherwise stated. PACU, postanesthesia care unit; SD, standard deviation; IQR, inter-quartile range.

effects for patients after the Nuss procedure [30]. Although the use of opioids is the most effective means of analgesia, excessive use may lead to various complications, including nausea/vomiting, constipation, urinary retention, and respiratory depression, as well as the potential for long-term opioid dependence [31]. Nerve blocks, such as paravertebral blocks and erector spinae plane blocks, can provide good postoperative analgesia for children undergoing the Nuss procedure as part of a multimodal analgesia strategy [18,19]. Peripheral nerve block combined with traditional local anesthetics, such as bupivacaine hydrochloride and ropivacaine hydrochloride, has a good analgesic effect in the first few hours after surgery, but rebound pain may occur as the peripheral nerve block subsides [8,9]. The main cause of rebound pain is the short analgesic duration of peripheral nerve block with traditional local anesthetics [8]. Hence, prolonging the duration of peripheral nerve block analgesia is the key to resolving postoperative rebound pain [8]. The development and application of ultralong-acting local anesthetics has become particularly important [32].

LB represents an extended-duration local anesthetic formulation with an established clinical history spanning multiple years [10]. In 2023, the U.S. Food and Drug Administration (FDA) approved expanded indications for LB, encompassing: (1) local infiltration analgesia for

Table 3

Comparison of primary and secondary outcomes between the liposomal bupivacaine (LB) and non-liposomal bupivacaine (nLB) groups.

Variable	LB(n = 54)	nLB(n = 55)	RR or MD(95 %CI)	P value
Primary outcome				
Postoperative morphine equivalent, median(IQR), mg				
Within 24 h (ITT analysis)	33.4 (18.8–50.9)	37.8 (24.6–56.0)	MD 4.4 (–1.6–13.2)	0.135
Within 24 h (PP analysis)	33.6 (18.8–51.2)	42.6 (24.6–60.3)	MD 8.9 (–1.6–15.7)	0.111
Within 48 h (ITT analysis)	60.9 (33.6–90.4)	74.5 (50.1–104.9)	MD 13.6 (0.04–27.9)	0.048
Within 48 h (PP analysis)	67.0 (34.7–91.1)	79.2 (53.8–108.6)	MD 12.2 (0.66–31.5)	0.043
Within 72 h (ITT analysis)	78.6 (44.2–110.2)	101.7 (61.5–137.8)	MD 23.1 (2.4–37.1)	0.023
Within 72 h (PP analysis)	86.0 (45.9–113.8)	109.4 (69.2–143.6)	MD 23.4 (3.1–42.3)	0.018
Secondary outcomes				
NRS maximum pain scores, median(IQR)				
In PACU	3(3–4)	3(3–5)	MD 0(0–1)	0.506
Day 1 after surgery	5(3–7)	6(3–7)	MD 1(0–1)	0.234
Day 2 after surgery	4(3–6)	4(3–6)	MD 0(0–1)	0.123
Day 3 after surgery	3(2–4)	3(3–4)	MD 0(0–1)	0.036
NRS minimum pain scores, median(IQR)				
In PACU	2(2–3)	2(2–3)	MD 0(0–0)	0.971
Day 1 after surgery	4(2–5)	4(3–5)	MD 0(0–1)	0.147
Day 2 after surgery	3(2–4)	4(3–4)	MD 1(0–1)	0.179
Day 3 after surgery	2(2–3)	3(2–4)	MD 1(0–1)	0.058
Moderate to severe pain	34(63.0)	43(78.2)	RR 0.81 (0.62–1.0)	0.081
Rebound pain	7(13.0)	16(29.1)	RR 0.45 (0.20–0.96)	0.039
NRS of sleep, median(IQR)				
Day 1 after surgery	5(4–6)	5(4–6)	MD 0(–1–0)	0.213
Day 2 after surgery	3(2–5)	4(3–5)	MD 1(0–1)	0.005
Day 3 after surgery	2(2–3)	3(2–4)	MD 1(0–1)	0.016
Postoperative nausea and vomiting	14(25.9)	12(21.8)	RR 1.2 (0.61–2.3)	0.615
Supplemental anti-emetics	7(13.0)	9(16.4)	RR 0.79 (0.33–1.9)	0.616
Time of first ambulation after operation, median (IQR),h	17.0 (14.0–21.0)	19.0 (15.0–23.0)	MD 2(0–4)	0.110

Data expressed as n(%) unless otherwise stated. ITT, intention to treat; MD, median difference; PP, per protocol; RR, relative risk; PACU, postanesthesia care unit; NRS, numerical rating scale; SD, standard deviation; IQR, inter-quartile range.

Table 4

Safety outcomes.

Variable	Participants, No. (%)		P value
	LB(n = 54)	nLB(n = 55)	
Hypotension	10(18.5)	9(16.4)	0.767
Intervention for hypotension	2(3.7)	2(3.6)	0.985
Hypertension	0(0)	1(1.8)	>0.999
Intervention for hypotension	0(0)	0(0)	>0.999
Bradycardia	0(0)	0(0)	>0.999
Tachycardia	2(3.7)	0(0)	0.243
Intervention for tachycardia	0(0)	0(0)	>0.999
Allergic reaction	3(5.6)	2(3.6)	0.679
Intervention for allergic reaction	1(1.9)	0(0)	0.495
Constipation	1(1.9)	0(0)	0.495
Numbness in the limb or skin	0(0)	0(0)	>0.999
Postoperative dizziness	6(11.1)	5(9.1)	0.726
Intervention for postoperative dizziness	0(0)	0(0)	>0.999
Postoperative headache	0(0)	1(1.8)	>0.999
Intervention for postoperative headache	0(0)	0(0)	>0.999

Data expressed as n(%).

patients aged ≥ 6 years; (2) interscalene brachial plexus blockade in adults; (3) popliteal sciatic nerve blockade in adults; and (4) adductor canal blockade in adults. Moreover, the use of LB in other nerve blocks has been reported in numerous studies. Nevertheless, existing studies demonstrate significant heterogeneity regarding LB's efficacy in postoperative pain management. Several studies have shown that LB has many benefits as a component of perioperative multimodal analgesia, such as reducing the intensity of postoperative pain and the amount of postoperative opioids used [11–14]. However, there are also studies showed that LB had no significant advantage in postoperative pain control [33,34]. Furthermore, a randomized controlled pharmacodynamics study in volunteers revealed that LB produced patterns of residual sensory blockade for up to 3.5 days, but these patterns were unpredictable in terms of intermittence, quality, and quantity [35]. A systematic review by Ilfeld et al. concluded that current evidence does not justify the routine clinical application of LB for postoperative analgesia and emphasized the need for large-scale, multicenter randomized controlled trials with rigorous methodology to validate its potential superiority over nLB [15]. In this randomized controlled trial, we found that paravertebral block with LB showed some advantages over paravertebral block with nLB in pain management after Nuss. These advantages are reflected in the saving of postoperative opioid consumption, the reduction of the incidence of postoperative rebound pain, and the improvement of postoperative sleep quality. Notably, rebound pain occurred in 7 patients in the LB group and 16 patients in the nLB group, with all cases emerging within the first 24 h postoperatively. Patients in the nLB group experienced rebound pain within 24 h after surgery, which was roughly the same as the time of resolution of the nerve block with non-liposome local anesthetics. However, rebound pain in the LB group also occurred within 24 h after surgery, which is inconsistent with the expected duration of analgesia with LB as an ultra-long-acting local anesthetic. As an ultra-long-acting local anesthetic, LB was designed to provide prolonged analgesia and nerve blockade. The time when rebound pain occurs in patients who received LB nerve block may be beyond 24 h after the surgery. Surprisingly, no rebound pain was documented in either group during the 24–72 h period. Potential explanations for the result include: (1) postoperative acute pain intensity is predominantly concentrated within the initial 24 h, with lower levels typically persisting during the 24–72-h period; and (2) the limited sample size of this study may have been underpowered to detect rebound pain occurrences later in the postoperative period. Further investigation is warranted to elucidate this phenomenon. Moreover, paravertebral block with LB did not show significant advantages in terms of pain score and incidence of moderate to severe pain after Nuss procedure. With respect to safety outcomes, paravertebral nerve block with LB did not result in a higher incidence of adverse events compared with nLB. Notably, in this study, the time used for paravertebral nerve block was approximately 30 min, which is comparable to the intercostal cryoablation procedure time. Thus, paravertebral block, like intercostal cryoablation, also prolonged the patient's operating room time.

Overall, while intercostal cryoablation, thoracic epidural analgesia, and paravertebral nerve block with LB each offer some advantages for post-Nuss procedure pain management, a comprehensive evaluation must weigh their respective limitations against cost-effectiveness considerations. Kim DH et al. found that interscalene nerve blocks with perineural LB provided effective analgesia similar to the perineural standard bupivacaine with dexamethasone after shoulder surgery [36]. There is no doubt that nLB combined with dexamethasone is more economical for patients than LB. However, whether paravertebral block with nLB plus dexamethasone provides similar analgesia to paravertebral block with LB in the Nuss procedure is not yet available and deserves further study. Moreover, the price of intercostal cryoablation for postoperative analgesia in our country is much higher than that of LB and nLB. However, there is currently no data to prove whether intercostal cryoablation for analgesia after Nuss surgery is superior to

paravertebral nerve block with LB. In the future, further research is still needed to evaluate the potential advantages of these techniques for postoperative analgesia in Nuss procedure from the perspective of clinical economics.

This study has several limitations. First, this single-center RCT focused on children scheduled for an elective Nuss procedure, and the generalizability and external validity of the findings are limited. Second, although LB used for paravertebral block significantly reduced opioid use after the Nuss procedure, the rate of moderate-to-severe pain and the NRS pain score remained high in each group. Moreover, large individual differences in pain intensity were noted. Further efforts and explorations are needed to find more effective or individualized analgesic regimens for these patients. Finally, because the paravertebral block was performed after the induction of anesthesia, we were unable to document the block range via a cutaneous sensory test.

5. Conclusion

In conclusion, we found that paravertebral administration of LB reduced opioid use after the Nuss procedure in children compared to paravertebral administration of nLB. Notably, although paravertebral administration of LB did not show a significant advantage in pain score and moderate-to-severe pain control, it significantly reduced the incidence of rebound pain after the Nuss procedure in children.

Contribution

Concept and design: LW, SQQ, ZD; Acquisition, analysis, or interpretation of data: SWW, ZC, LDJ; Drafting of the manuscript: LW, SWW, SQQ, ZD; Critical review of the manuscript for important intellectual content: SQQ, ZD.; Statistical analysis: LW, ZD; Obtained funding: SQQ, ZD; Administrative, technical, or material support: SQQ, ZD.

CRediT authorship contribution statement

Lei Wu: Writing – original draft, Software, Formal analysis, Data curation. **Si Wei Wei:** Data curation, Conceptualization. **Zheng Chen:** Data curation, Conceptualization. **Li Dan Jiang:** Data curation, Conceptualization. **Shuang Quan Qu:** Writing – review & editing, Project administration, Methodology, Funding acquisition, Conceptualization. **Zhen Du:** Writing – review & editing, Methodology, Funding acquisition, Data curation, Conceptualization.

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Declaration of competing interest

The authors declare no competing interests.

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